

Infection and Immune System

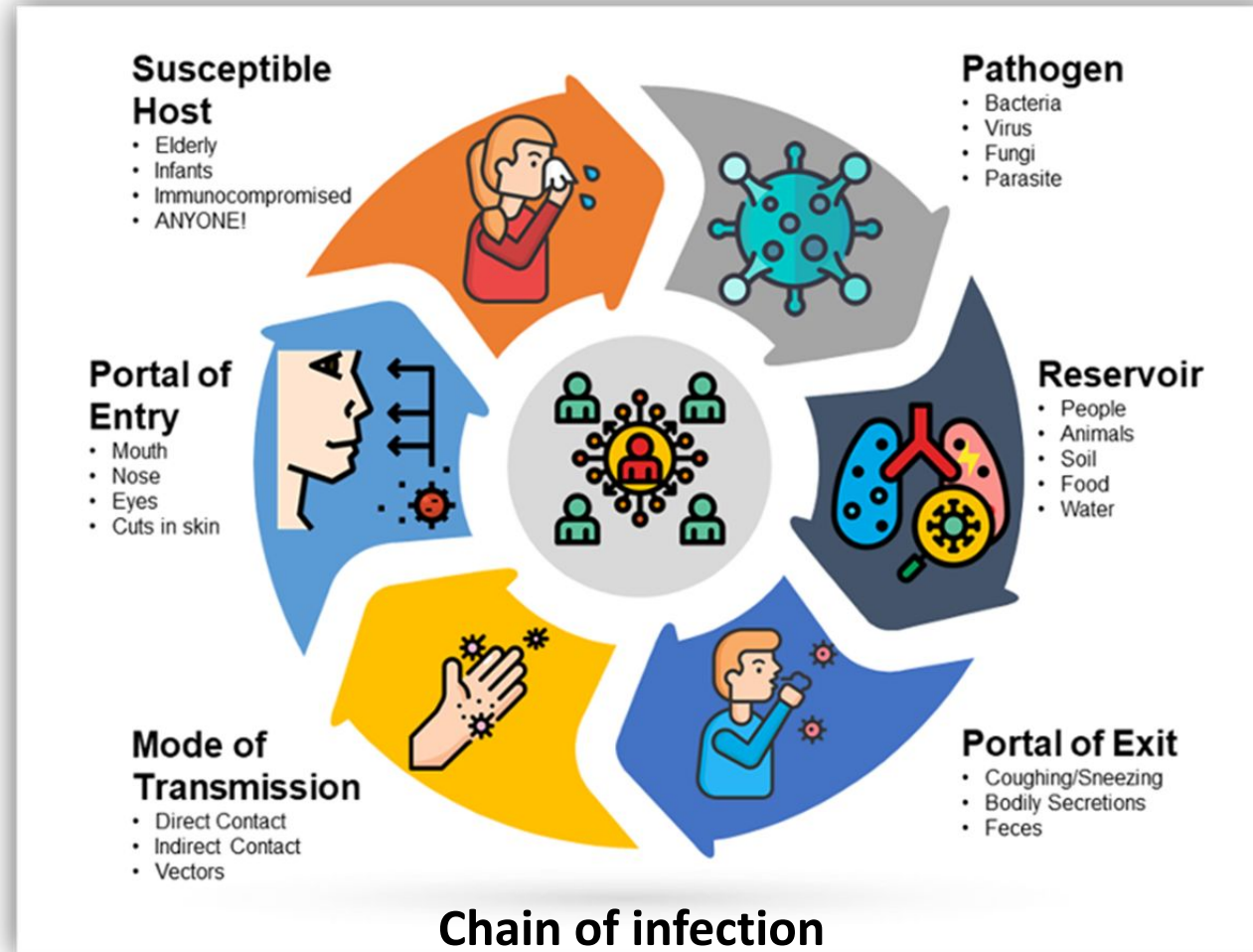
Introduction to Biology

Chapter: 10



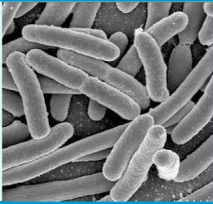
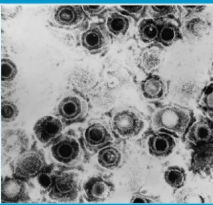
What is infection?

Infection is the invasion of a host organism's body tissues by disease causing agents (pathogens), multiplication of the pathogens and the reaction of the host against them.



Pathogens

- A pathogen is an organism, that can cause disease in its host.
- Pathogenicity refers to the ability of a pathogenic agent to cause disease.

Type of pathogen	Description	Human diseases caused by pathogens of that type
Bacteria <i>Escherichia coli</i> → 	Single-celled organisms without a nucleus	Strep throat, staph infections, tuberculosis, food poisoning, tetanus, pneumonia, syphilis
Viruses <i>Herpes simplex</i> → 	Non living particles that reproduce by taking over living cells	Common cold, flu, genital herpes, col sores, measles, AIDS, genital warts, chicken pox, small pox
Fungi <i>Death cap mushroom</i> → 	Simple organisms, including mushrooms and yeasts, that grow as single cells or thread like filaments.	Ringworm, athlete's foot, tineas, candidiasis, histoplasmosis, mushroom poisoning
Protozoa <i>Giardia lamblia</i> → 	Single-celled organism with a nucleus.	Malaria, "traveller's diarrhea" giardiasis, typano somiasis ("sleeping sickness")

Reservoir

Refers to humans, animals, plants, soil, water or inanimate matter in which a pathogen lives, multiplies, and depends for its survival.

Sources of infection:

1. Endogenous sources:

Normal flora that are usually non-pathogenic, resides with the host, but occasionally behave as pathogens.

2. Exogenous sources:

Pathogen enters the host's body from outside. For e.g.: humans, animals, insects, soil, water etc.

Reservoir

a. Humans: One of the major reservoirs!

- **Healthy carrier:** harbors the pathogen but never suffered from the disease caused.
- **Convalescent carrier:** one who has recovered from the disease but continues to harbor the pathogen in his body.
- **Temporary carrier:** this carrier state lasts for about 6 months.
- **Paradoxical carrier:** A person who acquires the pathogen from another carrier.
- **Contact carrier:** one who acquires the pathogen from a patient.

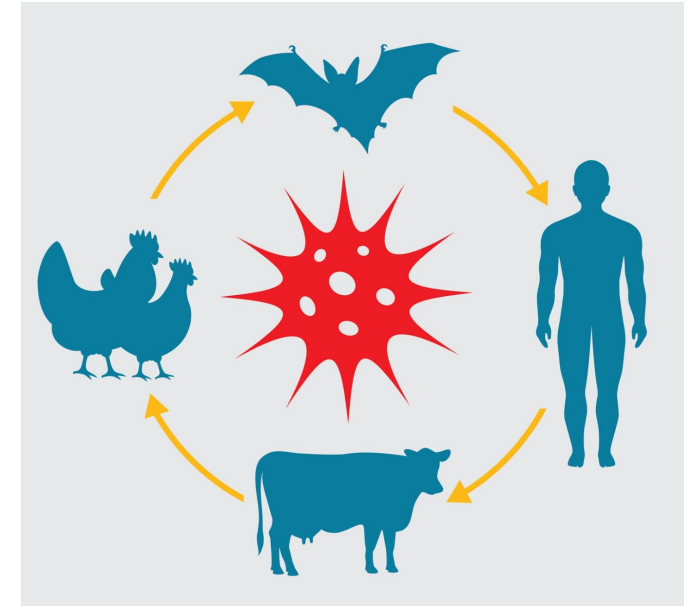
Reservoir

b. Animals: Infectious diseases transmitted from animals to humans are called zoonotic diseases.

These diseases can be: Bacterial (plague), Viral (rabies), Protozoal, Helminthic, Fungal

c. Insects: Diseases transmitted by insects are called arthropod-borne diseases. Insects transmitting these diseases are called vectors.

- **Mechanical vectors:** carry the pathogen on their wings, legs, and body. E.g. transmission of typhoid bacteria to humans through food by domestic flies.
- **Biological vectors:** the pathogen multiplies within the body of the vector. E.g. female anopheles mosquito harboring malarial parasite.



Reservoir

d. Soil and Water

- Some pathogens survive in the soil for long periods. E.g. tetanus spores, some fungi, parasites such as roundworm.
- Contamination of water with pathogens like *Vibrio cholerae* (cholera).



e. Food

- Contaminated food acts as a source of infection: food poisoning, diarrhea etc.



Modes Of Transmission

1. Contact

- a. Direct contact – directly through physical contact. STDs such as AIDS.
- b. Indirect contact- indirectly acquired through fomites (inanimate objects that can carry pathogens)

2. Airborne transmission

Droplets/ droplet nuclei (1-10micron diameter, remain airborne as aerosols) source of respiratory infections. Inhalation cause infection.

3. Ingestion

Intestinal infections like cholera, diarrhea etc. are acquired by ingestion of food or drink contaminated by pathogens.

Modes Of Transmission

4. Inoculation

Direct inoculation in the tissues, e.g. rabies virus inoculation through dog bite.

5. Transplacental

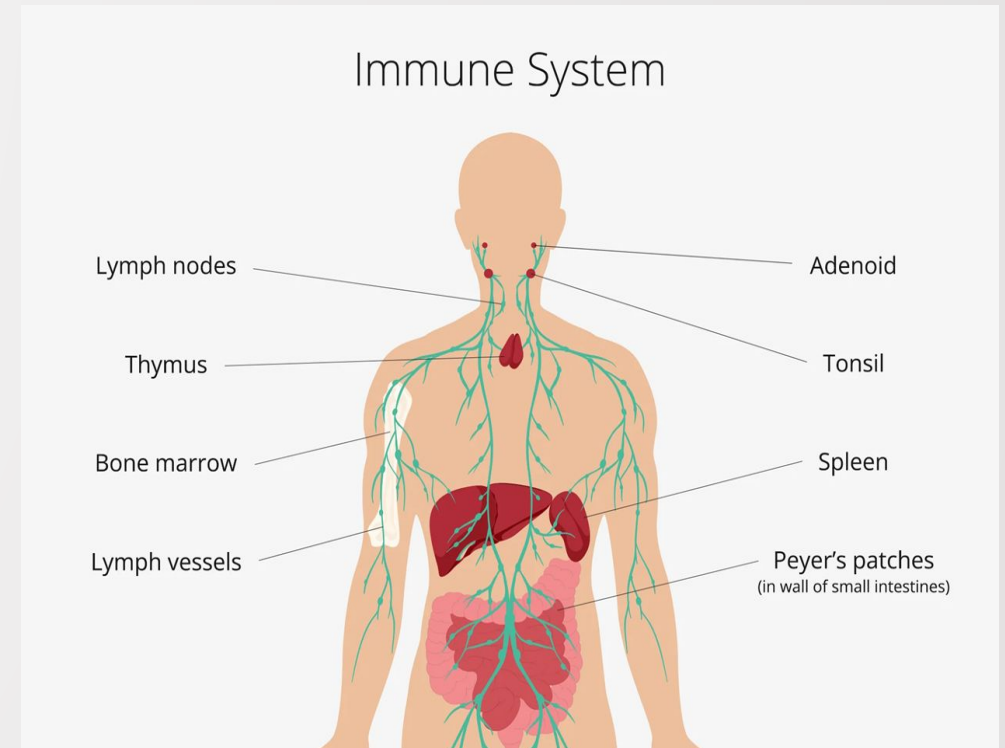
Transmission of the pathogen from mother to fetus via placenta. E.g. rubella virus.

6. Iatrogenic infection

Pathogen transmission during lab and surgical procedures such as blood transfusion, surgery. For e.g. AIDS, Hepatitis B.

Immune system: Major organs

- The immune system is a complex network of organs, cells and proteins that defends the body against infection, whilst protecting the body's own cells.
- Organs and tissues important to the proper functioning of the immune system include: the thymus, bone marrow, lymph nodes and lymphatic vessels, spleen, and skin.



1. **Bone Marrow:** All cells of the immune system begin their development here, from stem cells.
2. **Thymus:** Lymphocytes (WBC type) produced in bone marrow travel to thymus to become mature T cells.
3. **Liver:** Contains phagocytes (WBC type, ingest bacteria) and produce proteins of complement system.

Immune system: Major organs

4. **Spleen:** Collection of different types of WBC: B cells, T cells, and monocytes. A site for the immune system to interact with the pathogens.

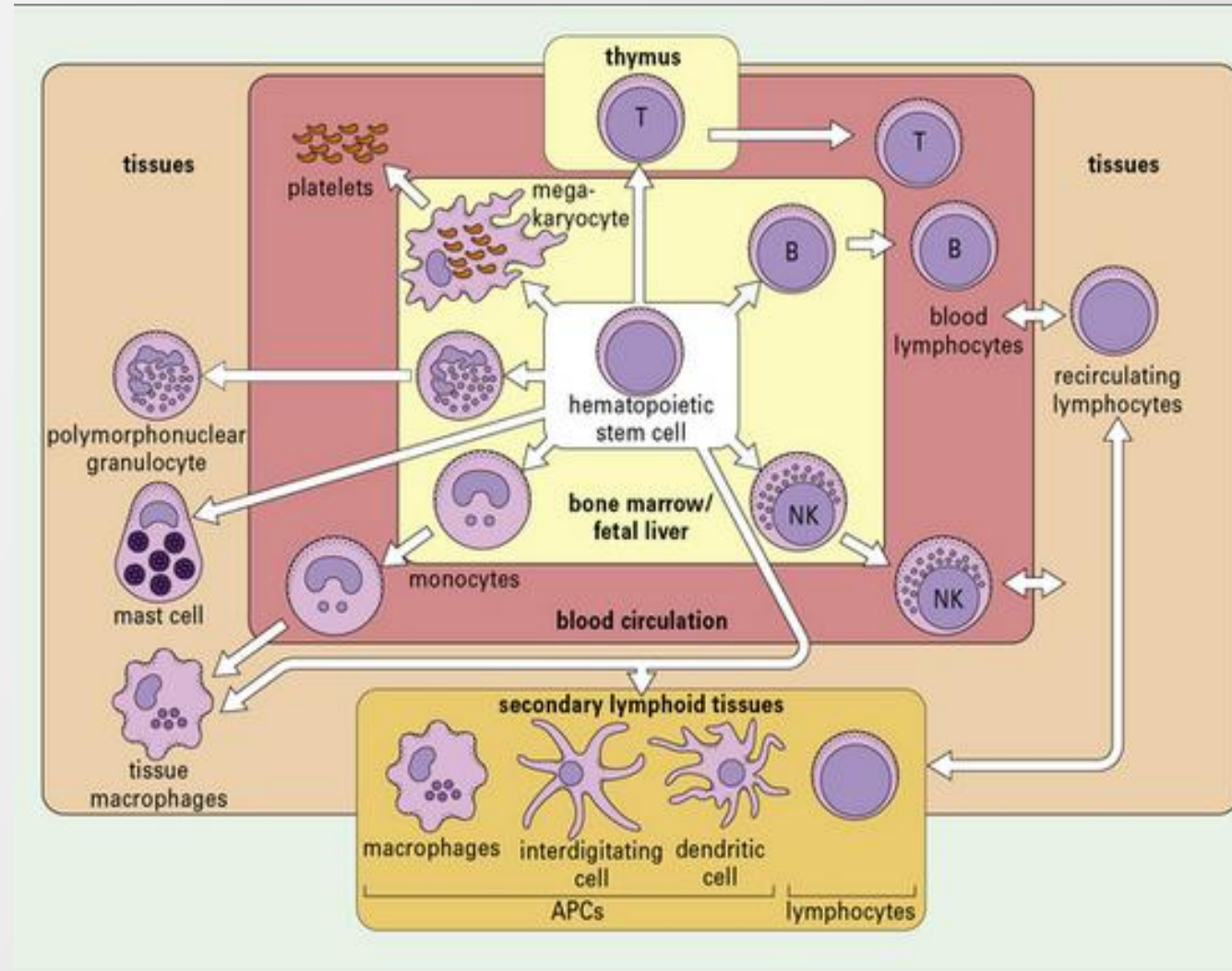
5. **Blood:** carries cells and proteins of the immune system from one part of the body to another.

6. **Lymph Nodes & lymphatic vessels:** Part of the lymphatic system. Lymph vessels carry lymph (excess tissue fluid) which is filtered through lymph nodes. Lymph nodes are collection of WBCs.

In the lymph nodes, the lymph is exposed to lymphocytes (WBC), pathogens are killed. The clean lymph and the lymphocytes move throughout the body via the vessels to fight infection.

Origin of cells of immune system

- All the cells of the immune system originate from the **hematopoietic stem cell** in bone marrow.
- **Platelets** released into bloodstream from **megakaryocytes**.
- Some cells pass through blood to tissue: **monocytes**
- Some cells found in all tissues: **mast cells**
- **B cells** mature in fetal liver/ bone marrow, **T cells** mature in thymus
- Some cells pass through lymph nodes: **lymphocytes**, **APCs** (antigen presenting cells)



Types of Immunity

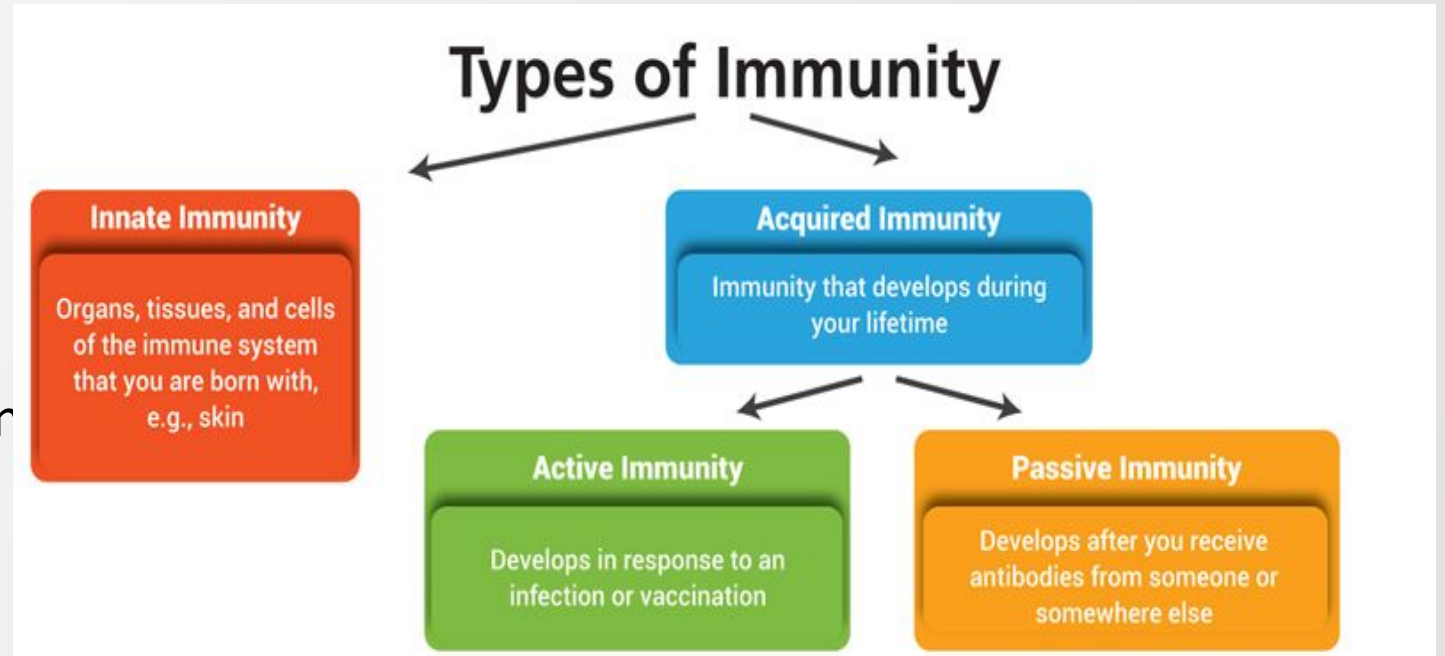
- In general, the immune system is divided into 2 categories:

1. Innate Immunity:

- nonspecific, ancient system
- not directed against any one pathogen but protect from all infections.

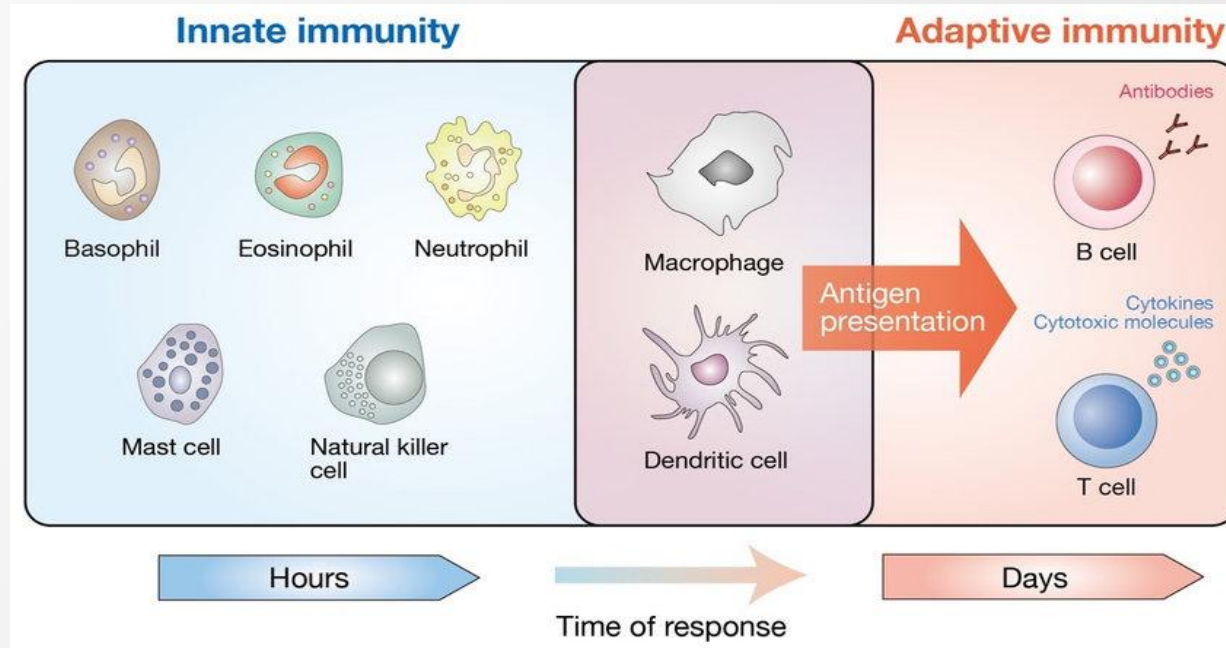
2. Adaptive/ Acquired Immunity:

- very specific, evolved later
- recognizes molecules produced by pathogens (antigens), attacks them and remembers them for future infections.



Cells and other components of the immune system can be divided into these two categories.

Cellular components of the mammalian immune system



- Cells of innate immunity provide the first line of defense within a short time after infection.
- As time passes, adaptive immunity starts to develop.
 - **B cells** give **humoral immunity** which is based on antibodies
 - **T cells** give **cell mediated immunity**: activated T cells gather other immune cells/ eliminate infected cells.
- **APCs** (part of both immunity): kill pathogens and present antigens (pathogen molecules) to B & T cells.

How does the immune system work?

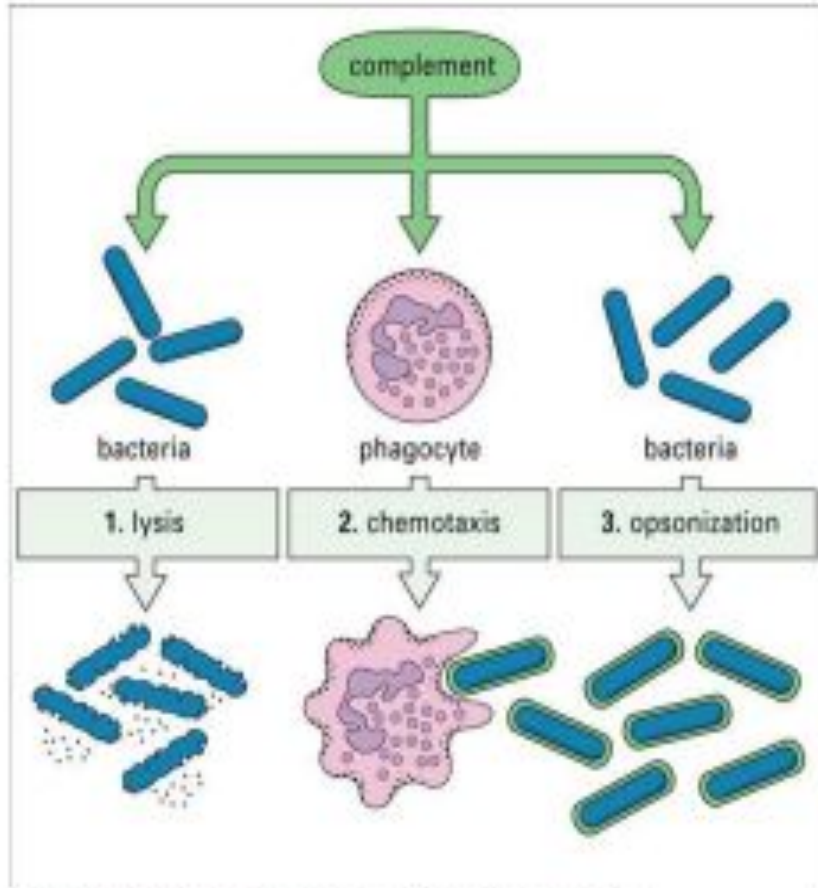
-Complement system

- Mostly part of **innate immunity**, can participate in **adaptive too**.
- Consists of proteins.

- **3 types of functions:**

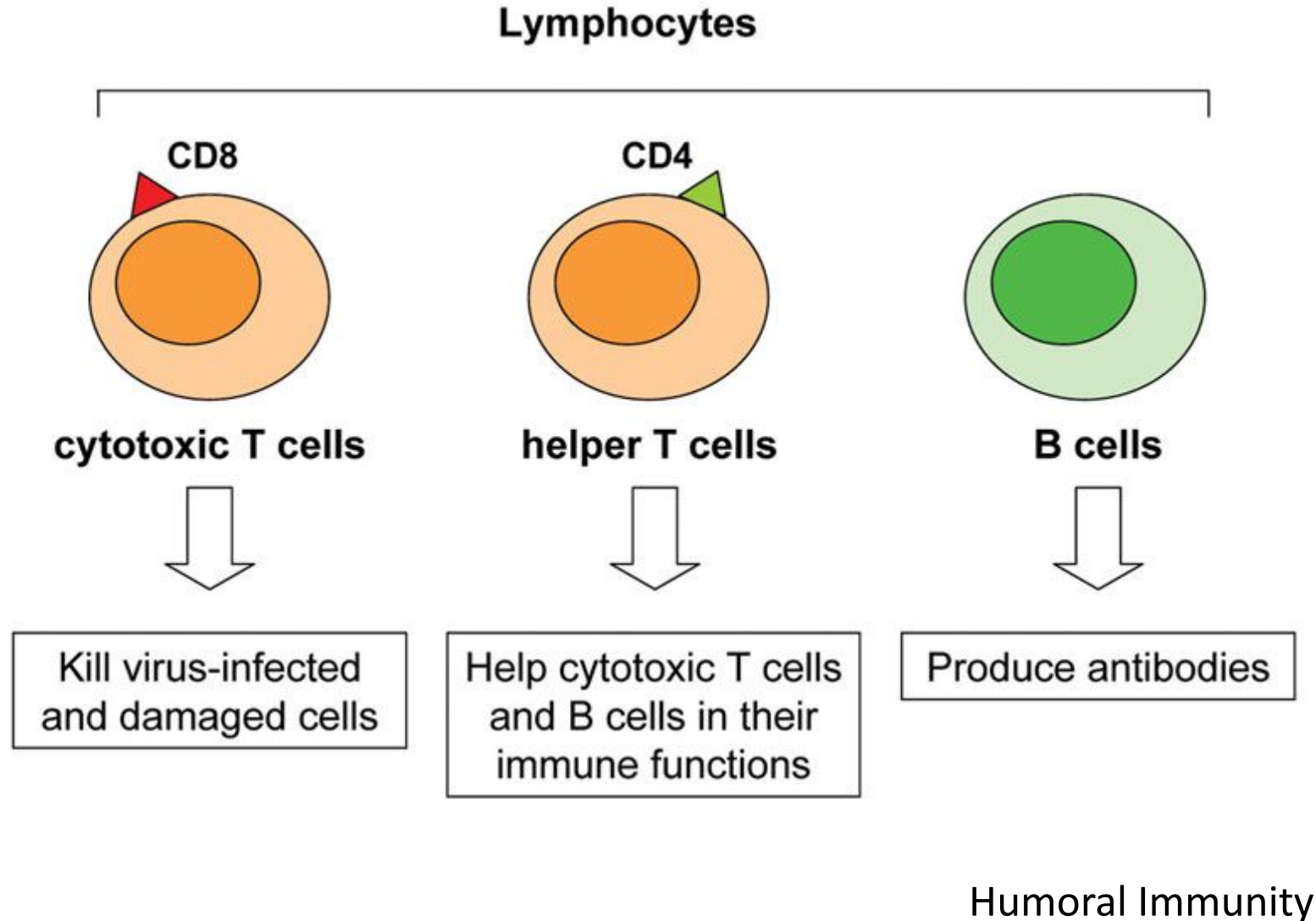
1. lyse the cell membrane of bacteria
2. attracts phagocytes (cells that can carry ingest and digest other cells) to the site of infection -> chemotaxis

3. coats bacteria with proteins which allows the phagocytes to recognize the bacteria -> opsonization



How does the immune system work?

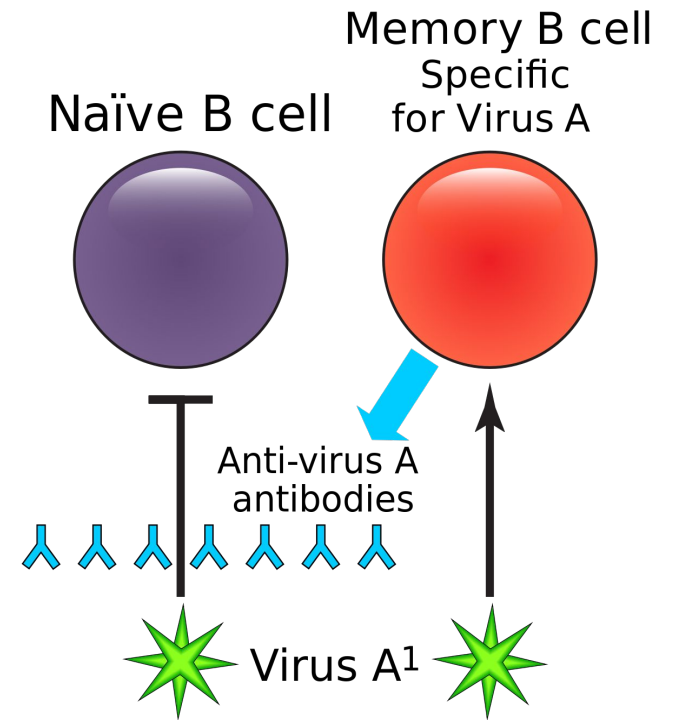
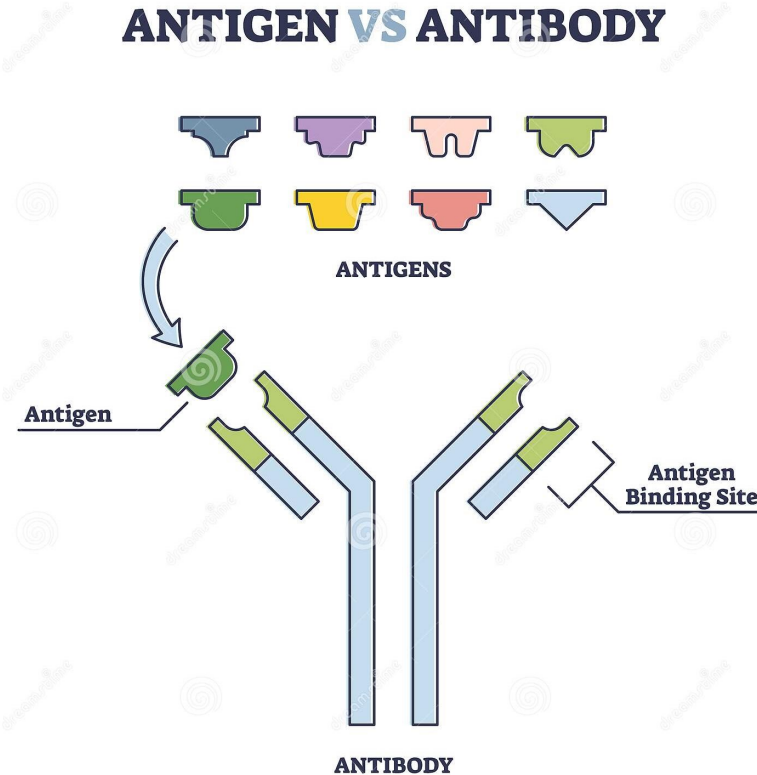
-Lymphocytes



How does the immune system work?

-Antibodies

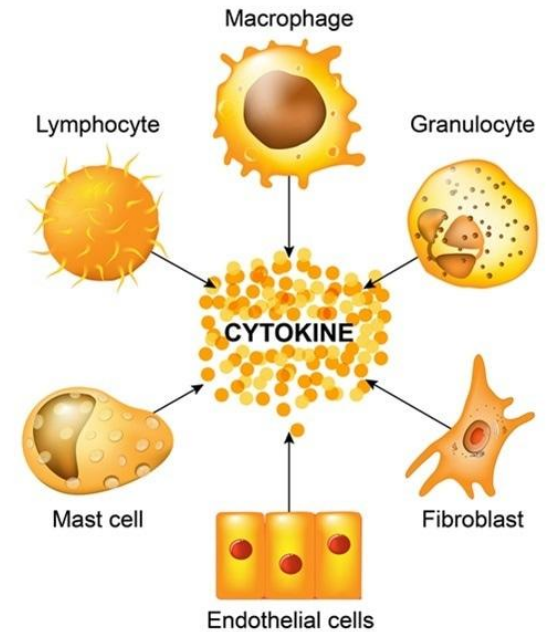
- **An antigen** of a pathogen/ foreign material is unwanted foreign substance that enters our body. Can be: bacteria, viruses, fungi, allergens, venom, toxins and disease-causing organisms.
- **An antibody** is a protein produced by the immune system to attack and fight off these antigens.
 - Part of **adaptive immunity**.



How does the immune system work?

-Cytokines

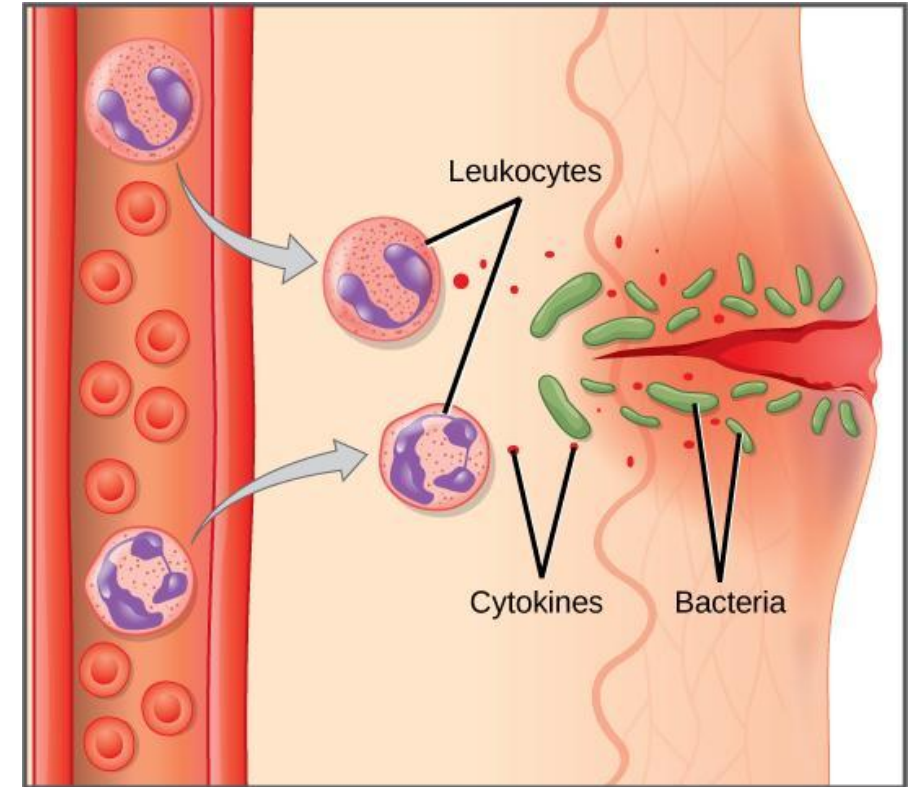
- Cytokines are mostly proteins produced by immune cells.
- Part of both **innate and adaptive immunity**.
- Have many functions but mostly carry out cell to cell communication to regulate the immune system. Serve as molecular messengers between cells.
- For e.g.
 - Interferon: regulate innate immunity
 - Interleukin: regulate growth and differentiation of immune cells
 - Tumor necrosis factor: induce inflammation, activates cytotoxic T cells



How does the immune system work?

-Inflammation

- The cells of the immune system are widely distributed throughout the body. But if an infection occurs it is necessary to concentrate them and their products to the site of infection. This is called inflammation.
- Part of **innate immunity**.
- What happens during:
 - Changes in blood flow
 - An increase in permeability of blood vessels
 - The migration of fluid, proteins, and white blood cells (leukocytes) from the circulation to the site of infection.



What if inflammation keeps going on?

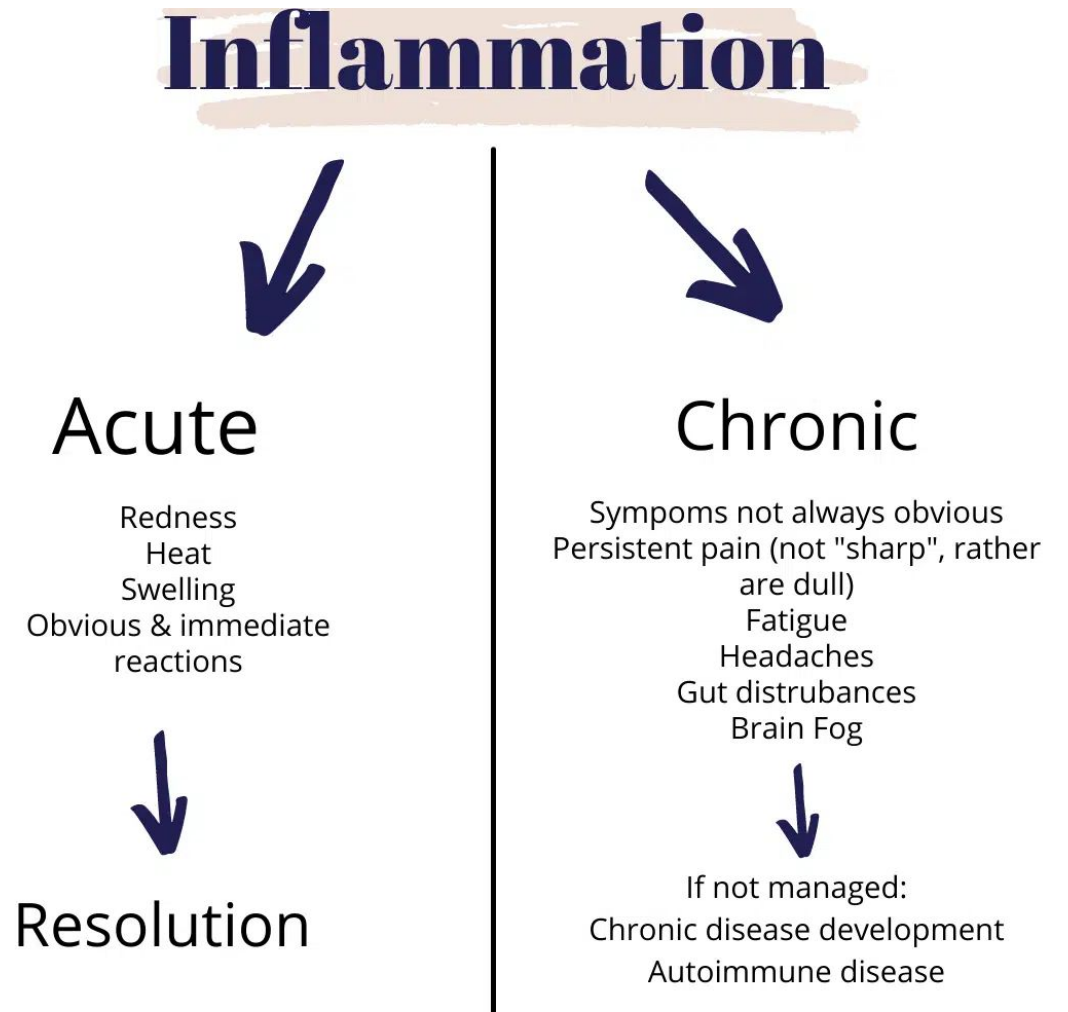
- Acute inflammation

happens immediately -> obvious symptoms -> resolved quickly

- Chronic inflammation

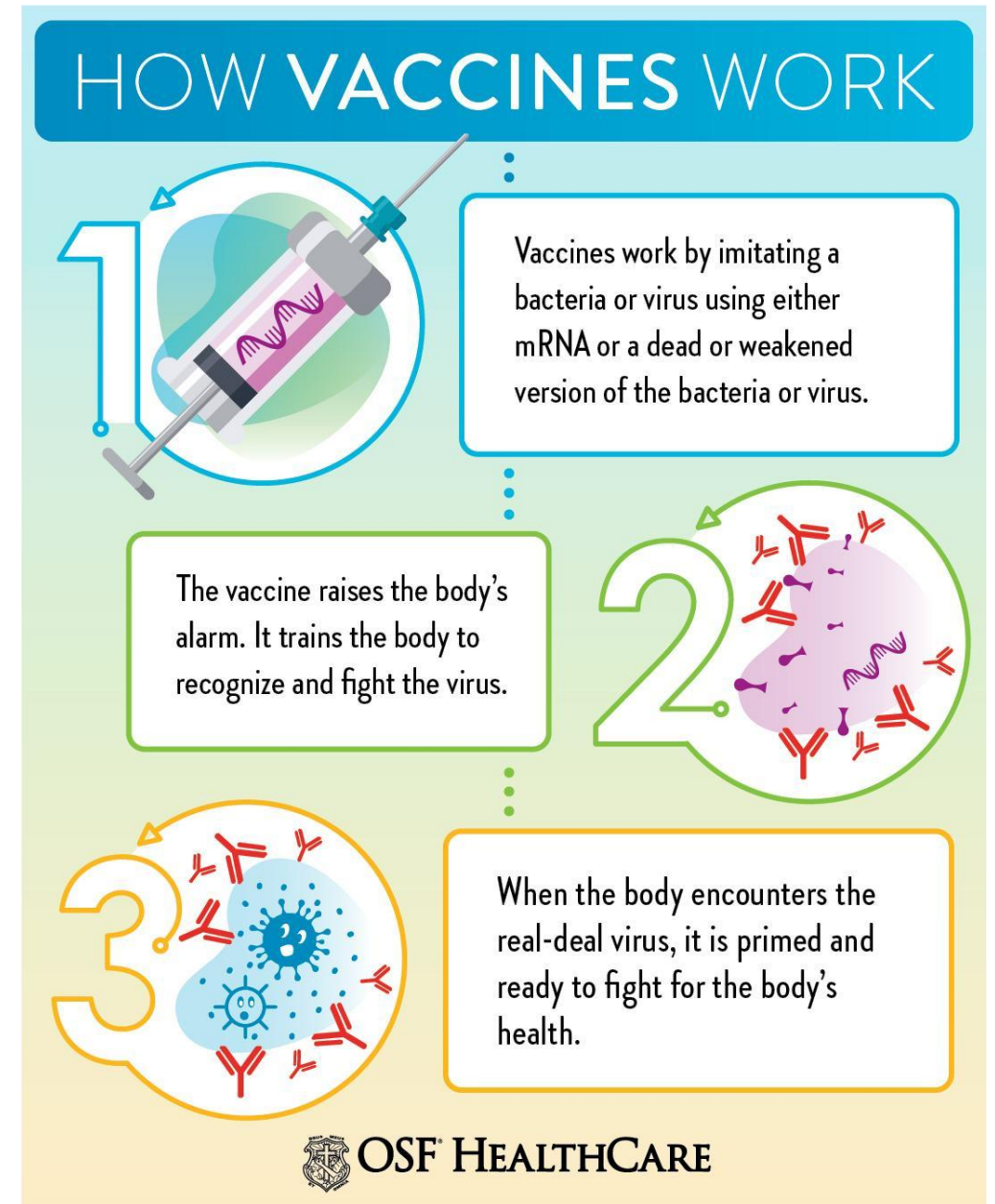
symptoms not obvious, dull -> takes time to go away

If not resolved -> can develop into diseases.



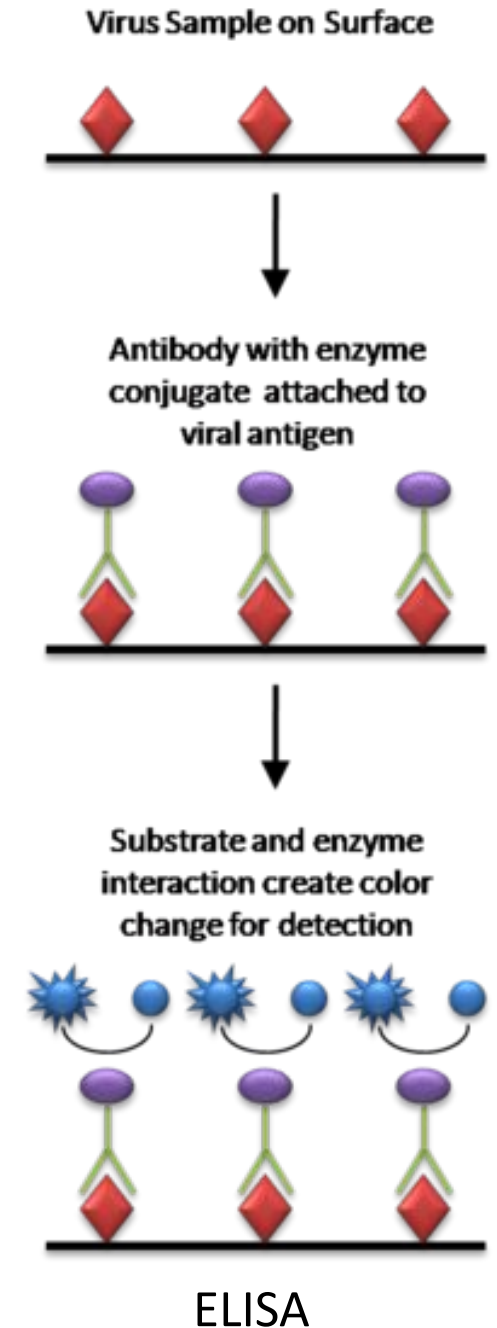
Vaccination

- Provide acquired immunity
- Works by taking advantage of antigen recognition and antibody response.
- mRNA/weakened/ killed form of the pathogen (antigen) is introduced into the body -> vaccine.
- The immune system responds, destroys and remembers it by producing specific antibodies.
- Initiate response when infected with the pathogen in future.



Diagnosis

- Different ways, an infection can be diagnosed:
 - **RT-PCR**: used to detect RNA viruses like coronavirus. RNA converted to DNA, DNA is amplified.
 - **Biosensors**: devices that converts biological response to recordable signals. Can detect cancer, viruses, glucose level.
 - **ELISA**: detects the presence of target antigens in a sample with the help of specific antibodies or vice versa. Can be used to detect COVID 19, HIV etc.



Thank you!

